

CS 255 Business Requirements Document Template

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Complete this template by replacing the bracketed text with the relevant information.

This template lays out all the different sections that you need to complete for Project One. Each section has guiding questions to prompt your thinking. These questions are meant to guide your initial responses to each area. You are encouraged to go beyond these questions using what you have learned in your readings. You will need to continually reference the interview transcript as you work to make sure that you are addressing your client's needs. There is no required length for the final document. Instead, the goal is to complete each section based on your client's needs.

Tip: You should respond in a bulleted list for each section. This will make your thoughts easier to reference when you move into the design phase for Project Two. One starter bullet has been provided for you in each section, but you will need to add more.

System Components and Design

Purpose

What is the purpose of this project? Who is the client and what do they want their system to be able to do?

- The Purpose is to design and implement a driver training management system for DriverPass to help students prepare for DMV driving tests.
- DriverPass is the client

- System should provide a platform for online classes, practice tests, and a scheduling interface for in-person driving lessons.
- Liam wants to fill a market gap using mix of digital content and hands-on training.

System Background

What does DriverPass want the system to do? What is the problem they want to fix? What are the different components needed for this system?

- There are high failure rates at the DMV and a lack of a centralized system to manage online coursework, physical driving reservations, and student progress tracking.
- System needs to be cloud based
- Accessible for both computers and mobile devices
- User Management Module: Handles different roles (Owner, IT, Secretary, Student) and permissions.
- Reservation: Manages 2-hour driving lesson slots, 10 cars, and assigned drivers.
- Educational Portal: Hosts online classes, practice tests, and progress tracking (Pass/Fail/In-Progress).
- Reporting Subsystem: Generates downloadable activity reports and internal tracking logs.

- Operational Constraints: Data can only be modified while online to prevent redundancy; the system must support automated password resets and secure credit card processing.

Objectives and Goals

What should this system be able to do when it is completed? What measurable tasks need to be included in the system design to achieve this?

- Enable students to register, select from three distinct training packages, and schedule/modify their own appointments online.
- Provide the Secretary with tools to manage appointments via phone/walk-in and allow the IT officer to manage user access and system security.
- Record specific lesson details: Start/End hour, driver name, and driver comments.
- Track student test status: Test name, time taken, score, and result.
- Log all transaction history
- Allow the owner to enable or disable specific training packages as business needs evolve
- Ensure the system is hosted in a way that handles backups and security automatically, minimizing technical downtime for the business staff.

Requirements

Nonfunctional Requirements

In this section, you will detail the different nonfunctional requirements for the DriverPass system. You will need to think about the different things that the system needs to function properly.

Performance Requirements

What environments (web-based, application, etc.) does this system need to run in? How fast should the system run? How often should the system be updated?

- The system must run on the web
- preferably cloud-based
- this will minimize the need for DriverPass to manage backups and security directly
- It should operate with minimal technical problems
- The system needs to be connected to the DMV to receive notifications for update regarding new rules, policies, or sample questions
- IT officer is responsible for maintaining and modifying the system

Platform Constraints

What platforms (Windows, Unix, etc.) should the system run on? Does the back end require any tools, such as a database, to support this application?

- The system must be accessible via any computer or mobile device with internet connectivity.

- So android,ios, windows, macos etc.
- Yes the back end does require a database to support the application (to track user records, activity logs, etc. to prevent data redundancy, for cloud hosting, and data export)

Accuracy and Precision

How will you distinguish between different users? Is the input case-sensitive? When should the system inform the admin of a problem?

- The system must enable tracking to identify which user made, canceled, or modified a reservation, and allow printing activity reports to assign responsibility.
- Yes input is case sensitive
- The system must allow the IT officer to block access for specific users, such as when an employee is let go.
- the system must clearly log who made, canceled, or modified a reservation.
- The system should automatically allow customers to reset their passwords if they forget them
- The system should notify DriverPass whenever the DMV has an update regarding new rules, policies, or sample questions.

Adaptability

Can you make changes to the user (add/remove/modify) without changing code? How will the system adapt to platform updates? What type of access does the IT admin need?

- The system should allow the owner to disable specific training packages without needing a developer to modify the code.
- The system needs to run off the web, preferably in the cloud, to minimize technical problems and allow the business to focus on operations.
- The system must allow data access from any computer or mobile device
- To handle compliance and updates, the system should connect to the DMV to receive notifications regarding new rules, policies, or sample questions
- The IT officer requires full access over all accounts
- This access level is necessary to maintain the system, reset passwords, and manage user permissions.

Security

What is required for the user to log in? How can you secure the connection or the data exchange between the client and the server? What should happen to the account if there is a “brute force” hacking attempt? What happens if the user forgets their password?

- The IT Officer requires full access to all accounts to reset passwords and block access for terminated employees
- Users need a unique username and a password to log into their accounts to access training materials and scheduling features

- developers must use HTTPS as the protocol to ensure that all data transfer, particularly sensitive information like passwords and financial details, is encrypted
- To mitigate brute force hacking attempts, the system must implement a maximum attempts policy for login. If the limit is exceeded, user accounts should be automatically locked for a designated amount of time, or require an administrator to unlock it

Functional Requirements

Using the information from the scenario, think about the different functions the system needs to provide. Each of your bullets should start with “The system shall . . .” For example, one functional requirement might be, “The system shall validate user credentials when logging in.”

- The system shall allow customers to create an account and make reservations for driving lessons online
- The system shall allow the secretary to schedule appointments via phone or in-person visits
- The system shall support three distinct training packages: Package One (6 hours), Package Two (8 hours plus in-person lesson), and Package Three (12 hours, in-person lesson, and online class access
- The system shall allow users to take online classes and practice tests
- The system shall track which customer is matched with a specific driver, time, and car

- The system shall provide a user interface displaying online test progress (tests taken, in progress, passed, failed) and driver notes/schedules
- The system shall notify DriverPass of updates to rules, policies, or sample questions from the DMV

User Interface

What are the needs of the interface? Who are the different users for this interface? What will each user need to be able to do through the interface? How will the user interact with the interface (mobile, browser, etc.)?

- The interface must include a form for filling in student information (name, address, phone, credit card details) and pickup/drop-off locations
- The interface must allow customers to view availability and make, cancel, or modify reservations for 2-hour driving lessons
- The system must include a way to contact the company and the students
- The main dashboard needs to display: online test progress
- The system must provide a page for contacting DriverPass and a way to contact students
- Customers need to be able to create accounts, make/modify/cancel reservations, take online classes, and take practice tests
- Secretary needs to be able to answer calls, enter customer info, schedule appointments
- Owner needs to be able to access data from anywhere to view progress and reports
- IT officer, needs full access to reset passwords and block terminated employee accounts

Assumptions

What things were not specifically addressed in your design above? What assumptions are you making in your design about the users or the technology they have?

- It is assumed that customers have access to a computer or mobile device to use the online feature
- It is assumed that the DMV provides a mechanism for connecting and pushing updates to the system

Limitations

Any system you build will naturally have limitations. What limitations do you see in your system design? What limitations do you have as far as resources, time, budget, or technology?

- Non-developers cannot easily add or remove entire modules, this requires a developer or system analyst
- Data cannot be modified or updated while offline to prevent data redundancy
- Time constraints (we have a deadline)
- We have a limited amount of developers

Gantt Chart

Please include a screenshot of the GANTT chart that you created with Lucidchart. Be sure to check that it meets the plan described by the characters in the interview.

